

Amman Verma
2401010141

DBMS - Assignment

Q:1 What is a Database Management System (DBMS)? Why is it require?

Ans: A DBMS is a software that allows user to create, store, retrieve, update and manage data in a structured way.

It acts as an interface b/w the user & the database ensuring that data is organized, secure and easily access.

Example: MySQL, Oracle Database, MongoDB.

Data Security: It control access to data through authentication & authorization mechanism.

Reduces data Redudancy: DBMS minimizes duplicates data to save storage space & min. consistency.

Data Integrity: It ensures accuracy & consistency of data using constraints, like primary key & foreign key.

Q:2 Why is DBMS better than traditional file system?

Ans: Less Data Redudancy: DBMS reduces duplicates data, while file system often store repeated copies.

Better Data Consistency: changes made in DBMS are reflected everywhere, unlike file system updation time.

Improved data Security: DBMS provide controll assess of permission where file systems have limited security control.

Q:3 What is Data Abstraction?

Ans: It is the process of hiding unnecessary details of data & showing only the essential information to users.

It helps reduces complexity & improves data security by restricting access to lower level details.

Physical Level

How data is actually stored in memory (files, indexes, storage)

Logical Level

Where data is stored in the database & the relationship b/w data

View Level

Shows only reserved data to user, hiding the rest of the database details.

Q:4 What is 3-Schema Architecture? Why is it important?

Ans: The 3 schema Architecture is a database design model that divides a database system into 3 levels to separate user application from physical database storage. It includes view-levels, logical levels, Physical level. It is important because it provides data independence, improves security by restricting user access, reduces system complexity.

Q:5 What is Data-Independence? Why is it needed?

Ans: It is the ability to modify the structure of a database at 1 level without affecting the structure at the next higher level. It is needed because it reduces system complexity, makes database maintenance easier, allows future modification without retraining application programs & improves flexibility.

Q:6 What are ER Models?

It is a diagram that represents their attributes, & relationship. It is used to database design planning.

Q: 7 Integrity Constraints

Ans: Rules that maintains data accuracy, & consistency in data.

Types:- Primary Key - Uniquely identifies each record & can't be null.

Foreign Key - Ensure value match a primary key in another table.

Unique - Ensures all values in a column are diff.

Not Null - Ensures a column can't have null values

Check - ensures column values satisfy a specified condition.

Section - B

Q: Draw three schemas architecture and explain each level.

External level
(user views / Applications)

⇒ shows how different users view specific parts of database

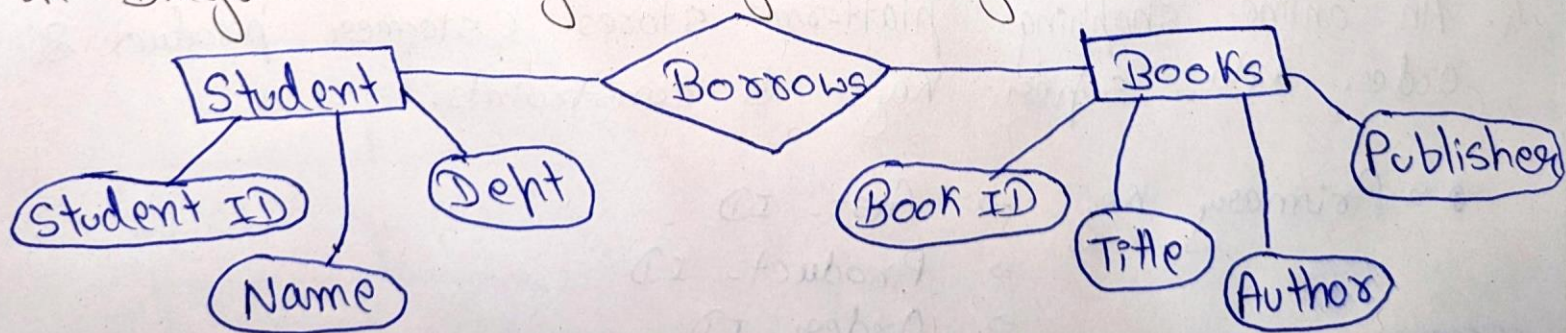
Conceptual level
(logical database schema)

⇒ Describes the overall logical structure of entire database.

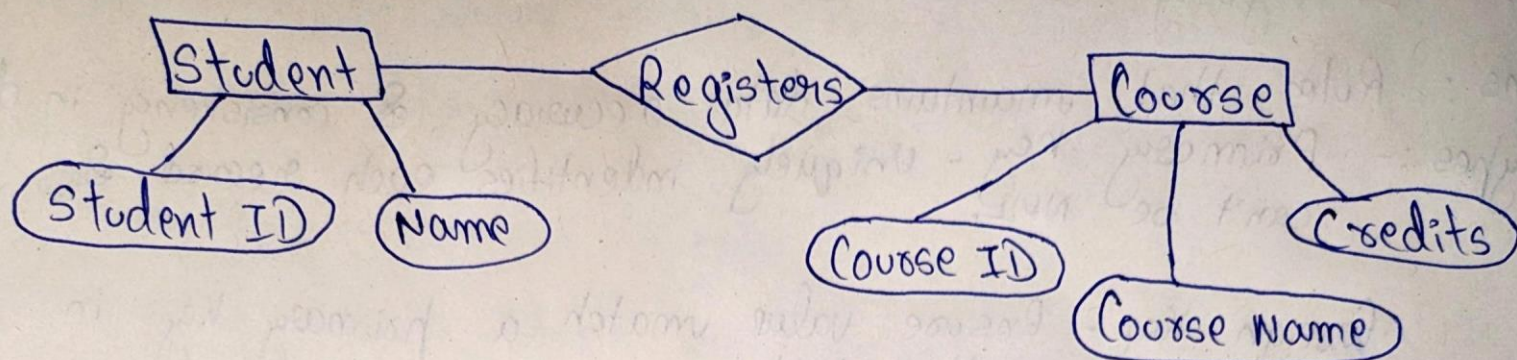
Internal level
(Physical storage level)

⇒ Describe how data is physically stored in memory.

Q: ER Diagram - Library management System



Q: ER Diagram - Student Course Registration



Section - C

Q: A hospital wants to store patients, doctors, & appointment data. Identify entities, attributes & relationship.

• Entities: Patient, Doctor, Appointment

Attributes

• Patient (Patient-ID, Name, Age, Gender)

• Doctor (Doctor-ID, Name, specialization)

• Appointment (Appointment-ID, Date, time)

Relationship

• Doctor treats Patients

• Patient books Appointment

Q: An online shopping platform stores customer, product & order data. Specify keys & constraints.

• Primary Key ⇒ Cust-ID
 ⇒ Product-ID
 ⇒ Order-ID

• Constraints ⇒ foreign keys

Not NULL

UNIQUE

check (price > 0)

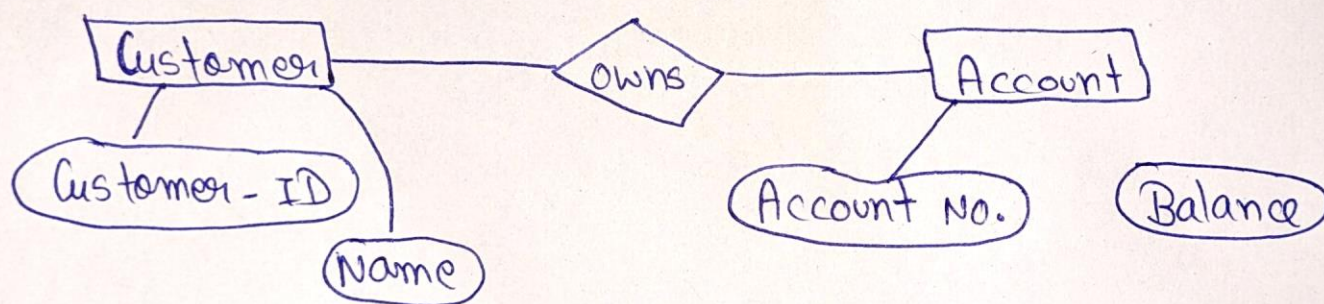
Q: A university stores student records. Explain how data abstraction helps different users.

- View level - student access
- logical view - Admin management
- Physical level - Data storage

Section D

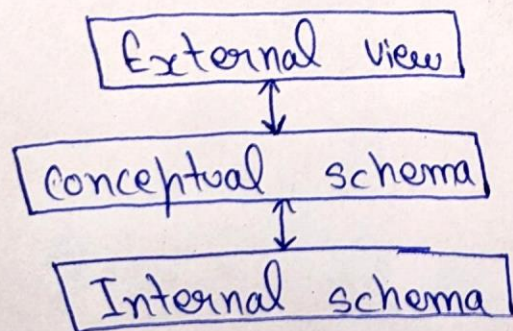
Q: Draw a simple ER diagram for any real-life system and explain any key design decision.

Ans: One customer can have multiple accounts.



Q: Draw the 3 schema diagrams architecture and mention one Benefit.

Ans:



Key Benefit: Data Independence
Physical changes don't affect user view, logical changes don't impact external application.